🎓 Professional Design & Visualization Course (16 Weeks)

🧭 Course Overview

This course is designed to make students **industry-ready** with practical skills in drafting, 3D modeling, rendering, and real project execution.

- Duration: **16 Weeks (4 Months)**
- Mode: **Offline / Studio-Based**
- Outcome: **Portfolio + Internship + Certification**

📅 WEEKLY STRUCTURE

🔷 Phase 1: AutoCAD 2D (Weeks 1–4)

Week 1 – Basics & Interface

- Introduction to AutoCAD
- Units, limits, layers, line types
- Drawing tools (Line, Circle, Arc, Rectangle)

Week 2 – Editing & Annotation

- Modify tools (Trim, Extend, Offset, Mirror)
- Text, Dimensions, Hatching

Week 3 – Working Drawings

- Floor plan drafting
- Section & elevation basics
- Electrical & plumbing layouts

Week 4 – Advanced & Printing

- Blocks & dynamic blocks

- Layouts & plotting
- 📐 **Design Class 1 (2D)**: Draft a complete residential plan

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◈ Phase 2: SketchUp 3D (Weeks 5–8)

Week 5 – 3D Basics

- Interface & navigation
- Push/Pull, groups, components

Week 6 – Modeling

- 3D house modeling
- Furniture & interior elements

Week 7 – Advanced Modeling

- Materials & textures
- Plugins introduction

Week 8 – Presentation

- Scene setup
- Export views
- 🏗️ **Design Class 2 (3D)**: Create full 3D model of building

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◈ Phase 3: Lumion Pro (Weeks 9–12)

Week 9 – Introduction to Lumion

- Import models from SketchUp
- Environment setup

Week 10 – Materials & Lighting

- Realistic materials
- Sun, sky, artificial lighting

Week 11 – Rendering & Animation

- Photo rendering

- Walkthrough animation

Week 12 – Advanced Effects

- Landscape, water, weather effects
- 🎥 **Design Class 3 (3D Visualization)**: Exterior + interior render

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◈ Phase 4: V-Ray Rendering (Weeks 13–16)

Week 13 – V-Ray Basics

- Lighting setup
- Camera settings

Week 14 – Materials & Textures

- Realistic materials (glass, wood, metal)

Week 15 – Interior Rendering

- HDRI lighting
- High-quality render output

Week 16 – Final Project

- 🏢 **Design Class 4 (Live Project)**:
  - Complete project (2D + 3D + Rendering)
  - Portfolio preparation

🎤 EXPERT LECTURES (4 Sessions)

1. **Architect (Site Planning & Design Thinking)**
2. **Civil Engineer (Execution & Drawings)**
3. **Interior Designer (Space Planning & Aesthetics)**
4. **3D Visualization Expert (Rendering Industry Insights)**

🏗️ DESIGN CLASSES SUMMARY

- ✅ 4 Classes – **2D Design (AutoCAD)**
- ✅ 4 Classes – **3D Design (SketchUp)**
- ✅ 4 Classes – **Live Project Design (Full Workflow)**

💼 INTERNSHIP PROGRAM

- **Minimum 1 Month Guaranteed Internship**
- Placement in:
  - Architecture firms
  - Construction companies
  - Interior design studios
- Work includes:
  - Site drawings
  - 3D modeling
  - Client presentations

🎯 FINAL OUTCOME

By the end of the course, students will have:

- ✔ Professional **2D drawings**
- ✔ Complete **3D models**
- ✔ High-quality **renders & animations**
- ✔ **Live project experience**
- ✔ **Internship certificate**
- ✔ Strong **portfolio for jobs/freelancing**

📜 CERTIFICATION

- Course Completion Certificate
- Internship Certificate
- Project Portfolio (Digital + Printable)